

Alkaline Phosphatase

Test Details

METHODOLOGY

Kinetic

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

Causes of high alkaline phosphatase include bone growth, healing fracture, acromegaly, osteogenic sarcoma, liver or bone metastases, leukemia, myelofibrosis, and rarely myeloma. Alkaline phosphatase is used as a tumor marker.^{3,4}

In rickets and osteomalacia, serum calcium and phosphorus are low to normal, and alkaline phosphatase may be normal or increased.

Hypervitaminosis D may cause elevations in alkaline phosphatase.

In Paget disease of bone there is often isolated elevation of serum alkaline phosphatase. Some of the highest levels of serum ALP are seen in Paget disease.

Hyperthyroidism, by its effects upon bone, may elevate alkaline phosphatase. There is evidence that thyroid hormone (T_3) acts to stimulate bone alkaline phosphatase activity through an osteoblast nuclear receptor-mediated process.⁵

Hyperparathyroidism, in some patients. Pseudohyperparathyroidism.

Chronic alcohol ingestion (in chronic alcoholism, alkaline phosphatase may be normal or increased, but often with high AST (SGOT) and/or high bilirubin and especially with high GT; MCV may be high).

Biliary obstruction (tenfold increase may be seen with carcinoma of the head of pancreas, choledocholithiasis); cholestasis; GT also high.

Cholecystitis with cholangitis. (In most patients with cholecystitis and cholangitis who do not have a common duct stone, alkaline phosphatase is within normal limits or only slightly increased.) Sclerosing cholangitis (eg, with ulcerative colitis), although importantly, 3% of cases of symptomatic sclerosing cholangitis may have normal serum ALP.⁶

Endoscopic retrograde cholangiography might be considered then in patients with diseases known to be associated with primary sclerosing cholangitis and with appropriate symptomatology even though ALP level is normal. Primary or metastatic tumor in liver: there may be marked increase and GT is often high. Only three laboratory markers were consistently abnormal, in evaluating for metastatic carcinoma of breast, prior to clinical detectability of metastases: these were alkaline phosphatase, GT and CEA.⁴

Cirrhosis, especially in primary biliary cirrhosis, in which fivefold or more increases are seen.

Gilbert syndrome: Increase in intestinal alkaline phosphatase is seen.⁷

Hepatitis: Moderate increases in alkaline phosphatase occur in viral hepatitis, but greater elevations of the transaminases (AST [SGOT], ALT [SGPT]) are usually found.

Fatty metamorphosis of liver (moderate increase occurs in acute fatty liver).

Diabetes mellitus, diabetic hepatic lipidosis.

Infiltrative liver diseases (eg, sarcoid, TB, amyloidosis, abscess).

Sepsis. Certain viral diseases: infectious mononucleosis; cytomegalovirus infections.

Postoperative cholestasis. Pancreatitis, carcinoma of pancreas, cystic fibrosis.

Pulmonary infarct (one to three weeks after embolism. Healing infarcts in other organs, including kidney, may also cause increased alkaline phosphatase); other situations in which angiofibroplasia occurs, such as healing in a large decubitus ulcer.

Tumors, especially hypernephroma; neoplastic ectopic production (Regan, Nagao isoenzymes).

Fanconi syndrome.

Peptic ulcer, erosion. Intestinal strangulation or obstruction, or ulcerative lesion. Steatorrhea, malabsorption (from bone, secondary to vitamin D deficiency). Ulcerative colitis with pericholangitis, other erosive lesions of colon.

Congestive heart failure.

Parenteral hyperalimentation of glucose, intravenous albumin administration.

Familial hyperphosphatasemia.

Idiopathic.

Drugs – estrogens (large doses), birth control agents, methyltestosterone, phenothiazines, oral hypoglycemic agents, erythromycin, or any drug producing hypersensitivity or toxic cholestasis. Many commonly and uncommonly used drugs elevate alkaline phosphatase, and tenfold increases may be seen with drug cholestasis.

Causes of low alkaline phosphatase are said to include: Hypothyroidism – but most hypothyroid patients have normal alkaline phosphatase.

Pernicious anemia – in very few patients.

Hypophosphatasia: Very low alkaline phosphatase values are found in the presence of normocalcemia or hypocalcemia. This diagnosis may be confirmed by quantitation of urinary phosphoethanolamine.

Malnutrition has been reported to relate to low values, but in practice, diseases causing malnutrition relate often to high alkaline phosphatase results (eg, disseminated neoplasia).

Some drugs (clofibrate, azathioprine, estrogens and estrogens in combination with androgens) lower serum ALP activity.

Special Instructions

Values obtained with different assay methods should not be used interchangeably in serial testing. It is recommended that only one assay method be used consistently to monitor each patient's course of therapy. This procedure does not provide serial monitoring; it is intended for one-time use only. If serial monitoring is required, please use the serial monitoring number **480004** (/tests/480004/alkaline-phosphatase-serial-monitor) to order.

Limitations

Used alone, alkaline phosphatase may be misleading.

Custom Additional Information

Serum alkaline phosphatase is a member of a family of zinc metalloprotein enzymes that function to split off a terminal phosphate group from an organic phosphate ester. This enzyme functions in an alkaline environment (optimum pH of 10). Active center of ALP enzymes includes a serine residue. Mg and Zn ions are required for minimal activity. Enzyme activity is localized in the brush border of the proximal convoluted tubule of the kidney, intestinal mucosal epithelial cells, hepatic sinusoidal membranes, vascular endothelial cells and osteoblasts of bone. There are distinctive forms of ALP in the placenta and small intestine; hepatic, renal and osteoblast (bone) ALP are similar molecules.

Serum ALP activity of intestinal origin occurs only in individuals of ABO blood type O or A. They are secretors of ABH RBC antigens and also carry the Lewis red cell antigen. Serum intestinal ALP level increases in these individuals about two hours following consumption of a fatty meal.

Liver alkaline phosphatase is increased in cholestasis and inflammatory liver disease as well as in infiltrative liver disease. The enzyme is sensitive to obstructive biliary processes, even small secondary bile duct obstruction, and thus may be increased in those patients when the bilirubin is normal due to compensatory bilirubin excretion by the rest of the liver. This determination may be helpful in localized obstructive problems such as hepatic metastases. An electrophoretically slow moving isoenzyme with high relative mass may occur in some patients with bile duct obstruction and hepatic metastases and may result in false elevation of CK-MB.⁷

To confirm biliary abnormality, an additional useful test is GT. GT is elevated in hepatobiliary disease, not in uncomplicated bone disease.

Serum ALP is increased during pregnancy. Marked decline of high ALP of pregnancy is seen with placental insufficiency and imminent fetal demise.

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

Separate serum or plasma from cells within 45 minutes of collection. If a red-top tube or green-top tube is used, transfer separated serum or plasma to a plastic transport tube.

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

See table.

Age^{1,2}	Male (IU/L)	Female (IU/L)
0 to 5 d	47–127	47–127
6 to 10 d	29–242	29–242
11 to 20 d	109–357	109–357
21 to 30 d	94–494	94–494
1 to 2 m	149–539	149–539
3 to 6 m	131–452	131–452
7 to 11 m	117–401	117–401
12 m to 6 y	158–369	158–369
7 to 12 y	150–409	150–409
13 y	156–435	78–227
14 y	114–375	64–161
15 y	88–279	56–134
16 y	74–207	51–121
17 y	63–161	47–113
18 to 20 y	51–125	42–106
>20 y	44–121	44–121

Storage Instructions

Room temperature

Causes for Rejection

Hemolysis; plasma specimen; specimen collected in EDTA tube; improper labeling

Footnotes

1. Pediatric reference intervals are based on Meites S. Pediatric Clinical Chemistry: Reference (Normal) Values. 3rd ed. Washington, DC: AACC Press; 1989:204.
2. Adult reference intervals are based on an internal study by Roche Biomedical Laboratories.
3. Narayanan S. Alkaline phosphatase as tumor marker. *Ann Clin Lab Sci*. 1983 Mar-Apr; 13(2):133-136. PubMed 6859804
(<http://www.ncbi.nlm.nih.gov/pubmed/6859804>)
4. Coombes RC, Powles TJ, Gazet JC, et al. Screening for metastases in breast cancer: An assessment of biochemical and physical methods. *Cancer*. 1981 Jul 15; 48(2):310-315. PubMed 6113047
(<http://www.ncbi.nlm.nih.gov/pubmed/6113047>)
5. Sato K, Han DC, Fujii Y, Tsushima T, Shizume K. Thyroid hormone stimulates alkaline phosphatase activity in cultured rat osteoblastic cells (ROS 17/2.8) through 3,5,3'-tri-iodo-L-thyronine nuclear receptors. *Endocrinology*. 1987 May; 120(5):1873-1881. PubMed 3569118
(<http://www.ncbi.nlm.nih.gov/pubmed/3569118>)
6. Lieverse AG, van Essen GG, Beukeveld GJ, et al. Familial increased serum intestinal alkaline phosphatase: A new variant associated with Gilbert's syndrome. *J Clin Pathol*. 1990 Feb; 43(2):125-128. PubMed 2318988 (<http://www.ncbi.nlm.nih.gov/pubmed/2318988>)
7. Butch AW, Goodnow TT, Brown WS, McClellan A, Kessler G, Scott MG. Stratus automated creatine kinase-MB assay evaluated: Identification and elimination of falsely increased results associated with high-molecular-

mass form of alkaline phosphatase. *Clin Chem.* 1989 Oct; 35(10):2048-2053. PubMed 2676240 (<http://www.ncbi.nlm.nih.gov/pubmed/2676240>)